GENAI INTERVIEW PREP

5 AI Engineer Interview Questions Most Candidates Get Wrong

RAG debugging · Multi-agent design · LLM evaluation
Scaling to 100K users · Healthcare AI safety

RAG | Agentic AI | LLM Evals | Production AI

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GenAI Engineer

RAG SYSTEM DESIGN

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Your company has built a RAG chatbot over 100,000 internal documents, but users report that it frequently hallucinates or misses relevant information. How would you debug and improve the system before considering a different LLM?

Answer 01

RAG SYSTEM DESIGN

I would first determine whether the issue originates from retrieval or generation. I'd evaluate the retrieval pipeline using metrics like Recall@K, Precision@K, and MRR to verify if relevant chunks are being retrieved. If retrieval is weak, I'd improve document chunking (prefer semantic or recursive chunking with appropriate overlap), evaluate a stronger embedding model, implement Hybrid Search (BM25 + Vector Search), and add a Cross-Encoder Re-ranker to improve relevance. Once retrieval is reliable, I'd refine the prompt to ensure the LLM answers only from the retrieved context, includes citations, and refuses to answer when evidence is insufficient. Finally, I'd benchmark the system using an evaluation dataset, measuring faithfulness, answer correctness, hallucination rate, latency, and cost. This structured approach isolates the root cause instead of assuming the model itself is the problem.

AI AGENT SYSTEM DESIGN

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You're building an AI customer support assistant that can answer FAQs, check order status, process refunds, and escalate complex cases. Would you build a single AI agent or a multi-agent system? Explain your design.

AI AGENT SYSTEM DESIGN

I would choose a multi-agent architecture because each task requires different capabilities and tools. A supervisor (or orchestrator) agent would first classify the user's intent and then route the request to specialized agents — for example, a RAG agent for FAQs, an API agent for order status, a transaction agent for refunds, and an escalation agent for handing conversations to human support. Each agent would have access only to the tools and permissions it requires, reducing unnecessary complexity and improving security. I'd also implement conversation memory, retries, fallback mechanisms, structured logging, monitoring, and human-in-the-loop workflows for sensitive actions. This modular architecture is easier to scale, debug, and maintain than a single monolithic agent.

LLM EVALUATION

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Your team wants to replace the current LLM with a newer model because it performs better on public benchmarks. As the AI Engineer, how would you decide whether to approve the migration?

Answer 03

LLM EVALUATION

I wouldn't rely solely on benchmark scores because they rarely reflect production workloads. Instead, I'd build an evaluation dataset using real user queries, edge cases, and business-critical scenarios. Both the existing and new models would be tested using the same prompts and evaluation criteria. I'd compare answer correctness, faithfulness, hallucination rate, latency, token usage, cost, and safety. For RAG applications, I'd separately evaluate retrieval metrics like Recall@K and Precision@K to ensure the retrieval pipeline isn't influencing the comparison. If the new model consistently outperforms the existing one without unacceptable increases in cost or latency, I'd validate it through shadow deployment or A/B testing before a gradual production rollout.

PRODUCTION AI SYSTEM DESIGN

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Your AI chatbot currently handles around 500 users per day, but the business expects traffic to increase to over 100,000 users daily. How would you redesign the system to handle this scale while maintaining performance?

PRODUCTION AI SYSTEM DESIGN

I would design the system as independently scalable services instead of a single application. The API layer, retrieval service, LLM inference, and monitoring would each scale separately. I'd use Redis to cache embeddings and frequently asked questions, optimize the vector database with indexing and metadata filtering, and stream LLM responses to reduce perceived latency. Load balancing and autoscaling would ensure the inference layer handles traffic spikes, while asynchronous processing would be used for non-critical tasks. I'd continuously monitor latency, throughput, token usage, infrastructure cost, and failure rates. Finally, I'd implement rate limiting, retries, circuit breakers, and graceful fallbacks to maintain reliability during peak traffic.

AI SAFETY & HALLUCINATION PREVENTION

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You're developing an AI assistant for the healthcare domain where incorrect information could have serious consequences. What techniques would you implement to minimize hallucinations and ensure trustworthy responses?

Answer 05

AI SAFETY & HALLUCINATION PREVENTION

In high-risk domains, I would ground every response using Retrieval-Augmented Generation with verified medical knowledge instead of relying solely on the LLM's internal knowledge. The prompt would explicitly instruct the model to answer only from the retrieved context, provide source citations, and decline questions when sufficient evidence isn't available. I'd implement input and output guardrails, confidence scoring, safety classifiers, and human review for high-risk recommendations. Continuous evaluation would measure faithfulness, hallucination rate, answer correctness, and compliance against a curated benchmark dataset. A production-grade healthcare AI should prioritize safety and transparency over answering every question, ensuring it knows when to defer rather than generate potentially harmful information.

Want more AI Engineer interview prep like this?

I break down real production GenAI problems — RAG, agents, evals, and system design — the way they actually get asked in interviews.

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Ritesh Rai — GenAI Engineer

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